

# The Wi-Fi didn't break. The login server did.

Eight passive sensors listened to one production hall for 30 minutes. Every client on the 802.1X network failed to log in, on every access point and every channel, while the IoT devices on the PSK network connected. Replaying the captures, Airframe sent the right alert to the right team 5 seconds after the first failed login.

|                                                         |                                                                |
|---------------------------------------------------------|----------------------------------------------------------------|
| <b>63 of 63</b><br>802.1X clients never logged in       | <b>5 s</b><br>from first failed login to alert                 |
| <b>18</b><br>devices kicked every 11–30 s from 14:30:53 | <b>0</b><br>radio faults found; the retry spike is traffic mix |

1,118,853 header-only frames, 8 sensors on 8 channels, 29 access points, 99 clients, 14:24–14:54 (Paris time). Device and network names are masked.

## Incident tape

One column per minute, all sensors merged. The red line is when the first alert fired; the amber line is the second.



## Findings, ranked

Each finding names what was observed, who owns it and the first check to run. Anything the captures cannot prove is said so.

alert 14:25:28

The APs only ever send identity requests and never start the login method. Every client the sensors could hear answered (129 of 129 attempts), so the APs are not getting a reply from the authentication server. After 3 tries, 30 s apart, the AP gives up with reason 23. No EAP-Success frame appears anywhere.

Owner: **identity / RADIUS team**    First check: **RADIUS logs and controller-to-RADIUS reachability**    Not proven: which component failed

alert 14:31:51

From 14:30:53 the APs tell 18 clients their session is no longer valid (reason 2), every 11–30 s, until the capture ends. Each client is always rejected by the same AP, and all 18 share one vendor prefix (Intel). None of them tries another AP, although other APs on the same network answer their probes.

Owner: **WLAN controller team**    First check: **controller events at 14:30:53, and why one device group**

## 14:26–14:28

Office devices' probe requests climb to 414 a minute at 14:27, in step with clients hitting their first login timeout (61 of 63 by 14:28): they are rescanning for a better AP. IoT devices rejoining in the same two minutes add a little more. The all-frame retry rate doubles to 9.5%, but probe responses are retried about 31% of the time in every minute. More of them in the mix raises the average; radio conditions did not change.

Owner: **none**    Action: **the RF team can stand down**

design

Most other channels carry 3. Everything on those two channels shares the same airtime, and beacons alone use more of it. A channel re-plan is a configuration change, not a hardware purchase.

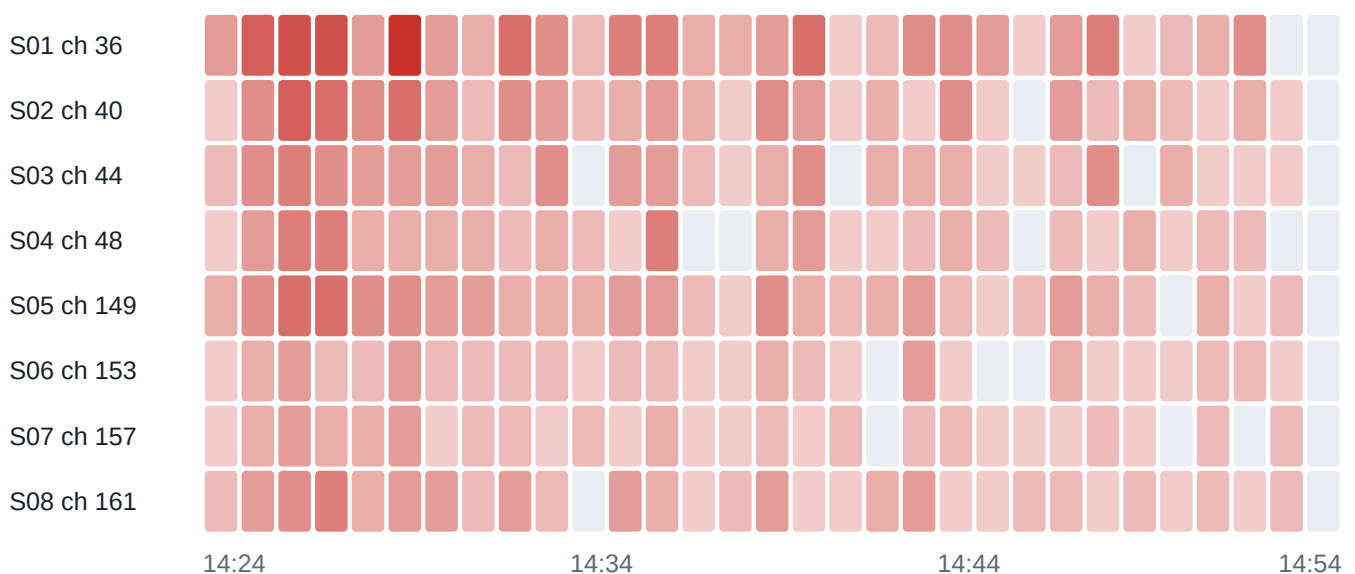
Owner: **WLAN design**

PSK network

All 31 successful joins in the capture were Raspberry Pi devices on the PSK network (23 devices). One of them stopped after key message 1 at 14:27 and did not retry: worth a look by the device team.

### Failed attempts by sensor and minute

Failures on all 8 channels in nearly every minute: the cause is shared, not local to one area.

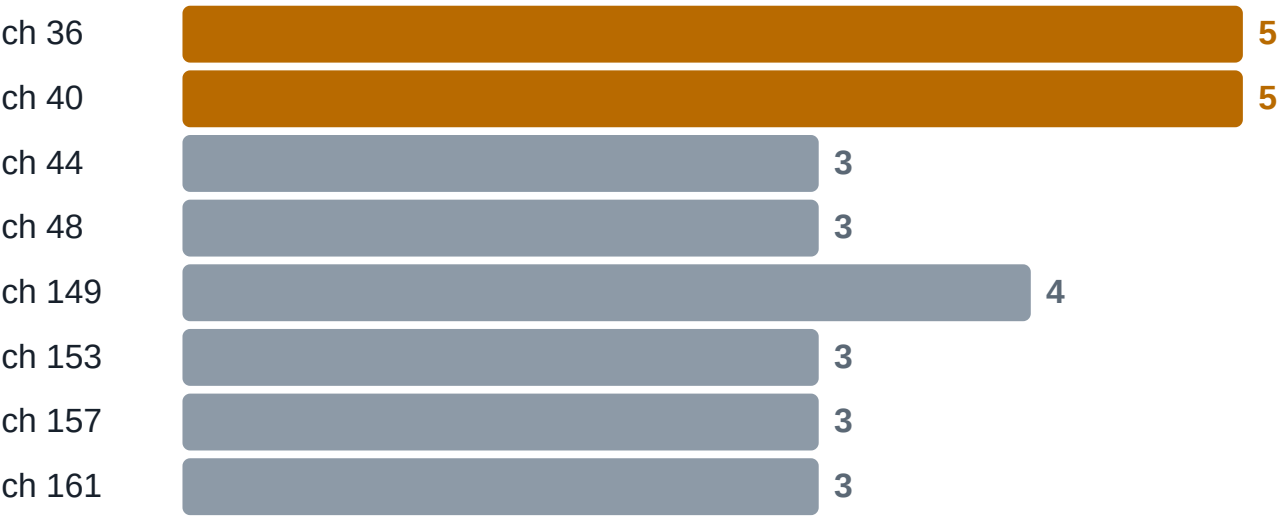


One client, one identity loop

Client C-155344 (masked). The AP asks for an identity every 30 s; the client answers each time; no login method ever starts; the AP disconnects it with reason 23 and it tries again.

| Time     | Direction    | Frame                       |
|----------|--------------|-----------------------------|
| 14:24:43 | client to AP | Authentication request      |
| 14:24:43 | AP to client | Authentication response     |
| 14:24:43 | client to AP | Association request         |
| 14:24:43 | AP to client | Association response        |
| 14:24:43 | AP to client | EAP Identity request        |
| 14:24:43 | client to AP | EAP Identity response       |
| 14:25:13 | AP to client | EAP Identity request        |
| 14:25:13 | client to AP | EAP Identity response       |
| 14:25:43 | AP to client | EAP Identity request        |
| 14:25:43 | AP to client | Deauthentication, reason 23 |
| 14:25:56 | client to AP | Authentication request      |
| 14:25:56 | AP to client | Authentication response     |
| 14:25:56 | client to AP | Association request         |
| 14:25:56 | AP to client | Association response        |

Access points per channel



What each sensor can hear

"AP side only" is the share of join attempts where the sensor heard the AP but never heard the client transmit. Sensors 01, 05 and 06 almost never hear clients directly: candidates for repositioning.

| Sensor   | Channel | APs | Clients in frames | AP side only |
|----------|---------|-----|-------------------|--------------|
| sensor01 | 36      | 5   | 37                | 100%         |
| sensor02 | 40      | 5   | 35                | 67%          |
| sensor03 | 44      | 3   | 31                | 72%          |
| sensor04 | 48      | 3   | 31                | 57%          |
| sensor05 | 149     | 4   | 34                | 99%          |
| sensor06 | 153     | 3   | 29                | 100%         |
| sensor07 | 157     | 3   | 32                | 46%          |
| sensor08 | 161     | 3   | 32                | 84%          |



# From “Wi-Fi is down” to the right team in seconds

Tesla already runs a fleet of passive sensors that capture 802.11 headers. Airframe turns those captures into ranked, owned incidents: what broke, where in the login sequence, how many devices, and which team should look first. No payloads, no transmitting, no new hardware.

## The same half hour, twice

The right column uses times measured from the captures. The left column is a typical response without Airframe, based on how Wi-Fi incidents usually unfold; it is illustrative, not measured.

| Without Airframe |  |                                                                                                |
|------------------|--|------------------------------------------------------------------------------------------------|
| 14:25            |  | Laptops and phones start failing to log in. Users retry, restart, walk to a different spot.    |
| 14:27            |  | Monitoring shows retries doubling. The RF team starts looking for interference: a red herring. |
| ~14:40           |  | Tickets reach the helpdesk: “Wi-Fi down in the hall.” The network team is paged.               |
| ~15:00+          |  | War room with RF, controller and identity teams. Someone eventually checks the RADIUS server.  |

| With Airframe |          |                                                                                                                                |
|---------------|----------|--------------------------------------------------------------------------------------------------------------------------------|
| 14:25:28      | measured | Critical alert to the identity team: logins stall at the Identity step on 26 APs; clients answer; check RADIUS.                |
| 14:28         | measured | The probe storm and retry spike are tagged as symptoms of the failed logins, not a radio fault. No reason to page the RF team. |
| 14:31:51      | measured | Serious alert to the controller team: 18 devices of one group kicked every 11–30 s since 14:30:53.                             |
| next          | outcome  | Two teams each work one clear problem, with the evidence attached and the example client to open.                              |

## Who uses it

### Wireless operations

Watches the incident board. Gets one merged incident per problem, not one alarm per sensor.

### Identity / RADIUS

Receives login-path alerts with the stage that failed and whether clients answered.

### Controller team

Receives disconnect waves with onset time, reason code, affected APs and device groups.

### Plant IT and production

Sees which device groups are affected, and whether production networks are healthy.

## Why it can be trusted

|                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Judgment built in</b><br>A roam is not a failure. One client heard by several sensors is one finding. A capture that ends mid-join is not a failure. Each rule has a test. |
| <b>Honest about blind spots</b><br>When a sensor hears only the AP, the finding says so. “Observed”, “consistent with” and “not proven” are kept apart in every alert.        |

**Private by design**

MAC addresses and network names are replaced with salted hashes before analysis. Login identities are never read.

# What faster answers are worth

Airframe saves money in one way: it shortens the time between “something broke” and “the right team is fixing the right thing”. The value of that time depends on who is offline. Move the sliders to match a site.

**What the captures cannot tell us:** there is no application data, so business impact is modelled, not measured. In this incident the affected devices were Intel and Apple clients on the 802.1X network, most likely laptops and phones; the IoT network kept working. The production-line figure is therefore off by default.

## This incident, as captured

|                                                                              |                                                                         |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <div>30 h</div> <div>device-hours locked out in 30 minutes</div>             | <div>6.4 h</div> <div>device-hours inside the rejection wave</div>      |
| <div>\$2,106</div> <div>of lost working time so far, at the rate below</div> | <div>\$35,280</div> <div>if it runs a full 8-hour shift unnoticed</div> |

## Per year

- Like this incident
- Some production devices affected
- Six sites

The incident

Users affected per incident 63

Measured here: 63 clients could not log in.

Loaded cost per user hour \$70

Assumption.

Time to detect (minutes)

Today 30 min

Assumption. Users usually notice before IT acts.

With Airframe 1 min

Measured alert: 5 s after the first failed login; 1 min allows for someone to read it.

Time to diagnose (minutes)

Today 60 min

Assumption. ZK Research: most Wi-Fi issues take over 30 min.

With Airframe 15 min

Assumption. The alert names the stage, owner and first check.

Engineering

Engineers pulled into triage 3

Assumption.

Loaded engineer cost per hour \$95

Assumption.

Incidents blamed on the wrong layer 38%

ZK Research: Wi-Fi blamed 64% of the time, root cause 40%, so ~38% of "Wi-Fi" incidents are something else.

Engineer hours lost per wrong-layer chase 4 h

Assumption.

Scale

Incidents per site per year 12

Assumption: one a month.

Sites 1

Production exposure

Share of incident time that stops a line 0%

Off by default: in this capture the IoT network kept working.

Cost of an idle line per hour \$2.30 M

Siemens, True Cost of Downtime 2024 (automotive).

Estimated savings per year

\$69,566

74 minutes saved per incident, worth \$5,797; 12 incidents a year.

|                            |          |
|----------------------------|----------|
| People back to work sooner | \$65,268 |
| Less triage time           | \$2,565  |
| No wrong-layer chases      | \$1,733  |
| Production protected       | \$0      |

Sensitivity: each 1% of incident time that stops a production line would add \$340,400 a year at these settings. That single assumption matters more than all others combined, which is why it starts at zero.

Compute cost is negligible by comparison: analysing all 8 sensors' 30 minutes takes about 4.5 seconds on one core, about 250,000 frames per second. At this site's traffic, one core keeps up with roughly 3,000 sensors.

Where the money comes from

People back to work sooner

Every minute cut from detection and diagnosis is a minute fewer for every affected user. Here, 63 people.

Less engineering time in war rooms

The alert names the owner and the first check, so fewer engineers spend less time triaging.

No chasing the wrong cause

In a 2016 ZK Research survey of 100 network administrators, Wi-Fi was reported as the cause 64% of the time but was the root cause only 40% of the time. Here the retry spike pointed at the radio; the real cause was the login server.

Production protected, when it applies

Siemens' True Cost of Downtime 2024 puts an idle automotive line at \$2.3 million per hour. If a login failure ever reaches scanners, robots or vehicles, each minute saved is worth about \$38,000 at full stop.

Sources: Siemens, *The True Cost of Downtime 2024* (automotive, \$2.3 M per idle line hour). ZK Research survey of 100 network administrators, 2016 (Wi-Fi blamed 64%, root cause 40%; most Wi-Fi issues take over 30 minutes to diagnose and resolve). All other defaults are assumptions and can be changed.

# How it works

Header-only captures in, ranked incidents out. The best of two independent implementations, cross-checked against each other and against 32 test captures.

## 1 Read and mask

Parse 802.11 and 802.1X headers from every sensor. Replace MACs and network names with salted hashes first.

## 2 Rebuild every join

Merge sensors per client and replay auth, association, 802.1X and the key handshake. Record where each attempt stopped and why.

## 3 Judge, then group

Separate roams, client exits and capture edges from real failures. Roll failures up into one incident per shared cause.

## 4 Route the alert

Send each incident to its owner with the evidence, the first check, a confidence level and what is not proven.

## Two independent tools, same answers

One implementation uses Wireshark’s dissector; the other parses frames from scratch. They agree on every headline number.

| Headline                             | From-scratch parser       | Wireshark-based |
|--------------------------------------|---------------------------|-----------------|
| 802.1X clients failing, APs affected | 63, 26                    | 63, 26          |
| Successful joins                     | 5 confirmed + 26 probable | 31              |
| Still in progress at capture end     | 13                        | 13              |
| Reason-2 disconnect frames           | 1,317 + 141 = 1,458       | 1,458           |
| Clients in the rejection wave        | 18                        | 18              |
| IoT handshake stuck at message 1     | 1                         | 1               |

## Tested beyond this dataset

### 16 real captures

Public Wireshark and aircrack-ng test captures: WEP, WPA2, PMF, WPA3, OWE, EAP-TLS and Suite-B, three header formats, broken timestamps and a disassociation flood.

### 16 scenarios with known answers

Generated multi-sensor captures: roam vs loop, the same event on two sensors, wrong passphrase, full AP, silent AP, congestion and a network-wide login failure.

### Privacy checked

For every scenario, the tests search all outputs for the raw MACs and network names used to generate it. None may appear.

## At scale

Sensors do the header extraction and send compact events, not captures. Each site correlates its own sensors by client, keeps summaries for everything and raw headers only around incidents. Alerts go to owners; a heartbeat from every sensor tells silent sensors apart from quiet air.

Data: 8 sensor captures, 14:24–14:54 local time. All device and network identifiers are masked. Business figures are estimates; see the Savings tab for sources and assumptions.